

Data Cleaning & Reporting Automation

Executive Pipeline Report | Generated on August 26, 2026 at 22:56:45

1. Executive Summary & Health Scorecard

Health Metric	Raw Ingestion	Cleaned Result	Pipeline Impact
Data Quality Score	75.1/100 [B (Acceptable)]	98.2/100 [A+ (Excellent)]	+23.1 Quality Gain
Total Rows	105	100	5 Duplicates Removed
Total Columns	9	16	Standardized snake_case
Missing Values	77	42	100% Imputed / Resolved

2. Key Performance Indicators (KPIs)

KPI Indicator	Computed Value	Description
Total Records	100	Total active rows in processed dataset
Unique Customers / Accounts	100	Distinct entities in 'emp_id'
Unique Products / SKUs	6	Distinct offerings in 'job_title'
Top Segment (Finance)	38 records	Most frequent group in 'department'

3. Visual Performance Analysis

Visual chart rendering skipped: name 'np' is not defined

4. Automated Insights & Findings

- **Insight:** Overall Data Health Score improved from 75.1/100 [B (Acceptable)] to 98.2/100 [A+ (Excellent)].
- **Insight:** Resolved 35 missing values across 6 columns using statistical imputation strategies.
- **Insight:** Eliminated 5 exact duplicate rows, preventing skewed metrics and double-counting.
- **Insight:** Successfully standardized data types for 2 columns (including currency symbol stripping, date parsing, and boolean normalization).
- **Insight:** Standardized 9 column headers into consistent snake_case format for SQL/pipeline compatibility.
- **Insight:** Top contributing category is 'Finance' accounting for 54.7% of total volume (\$12,690,000.00).
- **Insight:** Noticeable concentration risk: 'Finance' commands over half of the entire dataset revenue.
- **Insight:** Regional distribution is led by 'Remote' generating 65.3% of activity.

5. Strategic Recommendations

- **Action:** Implement frontend form validation at data entry to mandate required fields and eliminate downstream missing values.
- **Action:** Enforce database UNIQUE constraints on transactional ID fields to prevent duplicate records at ingestion.
- **Action:** Enforce strict schema typing at the ingestion pipeline to avoid storing raw currency symbols and unformatted strings in numeric fields.
- **Action:** Evaluate marketing and promotional strategies for lower-performing category 'Marketing' (5.5% share) to balance portfolio revenue.
- **Action:** Schedule automated daily/weekly runs of this cleaning pipeline to monitor data health drift over time.